

What are OSGi Web Applications?

This article covers OSGi concepts, the popular OSGi reference implementation called Equinox, and modular Web application development using the Equinox framework. Knowledge of Java-based Web application development would be sufficient to get the most out of this article.



Modular application development is a programming approach involving the assembly of distinct and loosely coupled modules to build composite applications. This approach has many benefits over monolithic application development. An important benefit is the reduced complexity, as it is easier to build and debug a small and independent unit. With loosely coupled modules, application development becomes agile. One module can be substituted by another in less time, and it leads to more creative solutions. Adding to that, the compile and build time of applications is reduced, as each module can be built independently. This will have a bigger impact in large-scale applications. By 'lazy loading' the modules, the application can be made to start 'in a flash'.

Due to these compelling benefits, modular application development is enticing to any developer. But there are a few challenges in implementing a module-based solution. Lack of elaborate language support is an important one. Many programming languages do not support modularity, out-of-

the-box. As a result, tooling has not matured extensively. This has led to low adoption rates and community support. OSGi fills these gaps in the Java arena, and hence it has become the *de facto* standard for modular development in Java. Though there is some serious effort going on in the Java community process (Java Development Kit) itself to add support for modularity, it is still in the nascent stages.

The history and the current state of OSGi

OSGi is a dynamic module system in Java. It is the outcome of an alliance formed between technology innovators who intended to create an open specification to build a modular application in Java. It was started as a home gateway framework in March 1999. This alliance works on the OSGi specification, and occasionally publishes it. The specification contains multiple chapters, each addressing different aspects of OSGi. There are various specification formats, such as core, compendium, enterprise, mobile and residential. The latest core specification is version